

Dialogue Summarization Using Pre-trained Language Models on the SAMSum Dataset

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Abstract

Automatic dialogue summarization condenses multi-turn conversational text into a concise, informative summary — a task of growing importance in customer service, meeting transcription, and social media analysis. In this paper, I investigate two complementary approaches on the SAMSum benchmark dataset, which contains 16,369 English messenger-style dialogues. First, I implement an extractive baseline using the TextRank graph algorithm, which selects key sentences without generating new text. Second, I fine-tune facebook/bart-large-cnn — a 406M-parameter BART model pre-trained on CNN/DailyMail news — on the SAMSum training split for 3 epochs using the HuggingFace Transformers framework. Experiments on 200 held-out test samples demonstrate that the fine-tuned BART model substantially outperforms TextRank, achieving ROUGE-1/ROUGE-2/ROUGE-L scores of 51.03/25.80/41.04 compared to 28.72/7.94/22.20. The most notable gain is in ROUGE-2 (+17.86, a 3.25× relative improvement), confirming that abstractive models capture inter-turn semantic relationships that extractive methods fundamentally cannot. Qualitative case study analysis identifies both success patterns and systematic failure modes, including hallucination on very short dialogues and over-generalization on informal slang-heavy conversations.

1. Background Survey

1.1 History of Automatic Text Summarization

Automatic text summarization has been studied since the 1950s. Luhn [1] proposed one of the earliest methods based on word frequency, selecting sentences that contained the most frequent non-trivial words. Edmundson [2] extended this with additional heuristics including sentence position and cue words. These early methods are now classified as extractive summarization: they select and return sentences verbatim from the source without generating new text, making them computationally lightweight and factually safe but limited in expressiveness.

The transition to neural methods began with Rush et al. [3] in 2015, who applied an attentional encoder-decoder to headline generation on the DUC benchmark, producing the first fully abstractive neural summaries. See et al. [4] introduced Pointer-Generator Networks in 2017, combining copying from the source with free generation, which substantially reduced hallucination — a critical failure mode where models generate plausible-sounding but factually incorrect content.

The transformer revolution [5] catalyzed modern summarization. BERT [6] established

bidirectional pretraining as a powerful representation learning paradigm. BART [7] and T5 [8] demonstrated that sequence-to-sequence pretraining on large noisy corpora yields powerful generation capabilities transferable to downstream summarization. PEGASUS [9] further refined this by designing a summarization-specific pretraining objective: masking complete sentences that are most likely to appear in gold summaries, yielding state-of-the-art performance on multiple benchmarks with relatively little fine-tuning data.

1.2 Dialogue Summarization and the SAMSum Dataset

While news summarization has been extensively studied, dialogue summarization presents unique challenges. Gliwa et al. [10] identified the key difficulties: (1) multi-speaker structure requires capturing inter-turn interactions, not just individual utterance content; (2) informal language including abbreviations, emoji, and colloquialisms differs markedly from formal written text; (3) topics drift across turns, requiring models to identify the main conversational thread; and (4) implicit co-references demand cross-turn reasoning. To address the lack of training data, Gliwa et al. introduced the SAMSum Corpus, consisting of 16,369 messenger-like dialogues created by linguists and paired with human-written abstractive summaries. It rapidly became the standard benchmark for dialogue summarization.

Subsequent work has proposed dialogue-aware model variants. Feng et al. [11] introduced topic-aware contrastive learning to improve summary coherence. Liu et al. [12] proposed discourse graph-based encoders that explicitly model speaker turn relationships. More recently, large language models (LLMs) such as GPT-4 have been applied to dialogue summarization in zero-shot and few-shot settings, achieving competitive performance without task-specific fine-tuning, though at considerably higher inference cost.

1.3 Limitations of Current Approaches

Despite strong ROUGE scores, important limitations remain. First, ROUGE is a recall-oriented n-gram overlap metric that correlates imperfectly with human judgments of fluency and factual faithfulness [13]. Second, transformer models are constrained by maximum input length (typically 512–1024 tokens), forcing truncation of longer dialogues and potential loss of important context. Third, models fine-tuned on SAMSum — which covers casual everyday messaging — tend to generalize poorly to domain-specific conversations (medical consultations, legal proceedings, technical support) without additional adaptation. These limitations motivate our experimental focus on quantitative ROUGE evaluation combined with qualitative case study analysis to understand where the model succeeds and fails.

2. Methodology

2.1 Dataset: SAMSum

I use the SAMSum Corpus [10] loaded from the HuggingFace Datasets library (knkarthick/samsum). The dataset contains 16,369 English messenger-style dialogues split into train, validation, and test sets. Table 1 presents the key statistics obtained from our data exploration cell.

Split	Samples	Avg. dialogue (words)	Avg. summary (words)
Train	14,731	93.8	20.3
Validation	818	91.6	20.3
Test	819	95.5	20.0

Table 1: SAMSum dataset statistics.

The average compression ratio is 78.3% on the training set, meaning each summary retains only about one-fifth the length of its source dialogue. This aggressive compression, combined with the informal conversational register, makes SAMSum substantially more challenging than news summarization benchmarks such as CNN/DailyMail.

Figure 1 shows the word-count distributions for dialogues and summaries in the training set. Dialogues follow a right-skewed distribution with a long tail extending to 803 words, while summaries are tightly concentrated between 10 and 40 words. This disparity underscores the importance of strong compression capability in the summarization model.

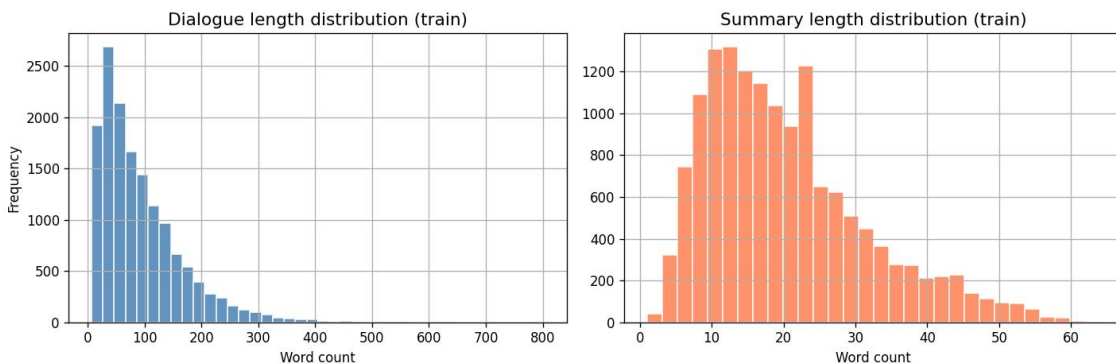


Figure 1: Word count distributions for dialogues (left) and summaries (right) in the SAMSum training set.

2.2 Baseline: TextRank Extractive Summarization

As a non-neural baseline I apply the TextRank algorithm [14], implemented via the Sumy library. TextRank represents sentences as graph nodes and computes edge weights based on word overlap between sentence pairs. Iterative PageRank-style computation assigns an importance score to each sentence, and the top-2 highest-ranked sentences are concatenated as the summary. TextRank requires no training data and runs in milliseconds per sample, but is fundamentally limited: it copies sentences verbatim and cannot synthesize information across multiple turns.

2.3 Model: BART Fine-tuning

My primary model is **facebook/bart-large-cnn**, a 406.3M-parameter BART [7] encoder-decoder model pre-trained on large English corpora and further adapted to news summarization on CNN/DailyMail. I fine-tune this model on the SAMSum training set for 3 epochs using the HuggingFace Seq2SeqTrainer. The dialogue is prefixed with "summarize: " to reinforce the task. Tokenization truncates inputs to 512 tokens and targets to 128 tokens. Training uses the

AdamW optimizer with FP16 mixed precision on GPU.

Hyperparameter	Value
Base model	facebook/bart-large-cnn
Parameters	406.3M
Epochs	3
Learning rate	5×10^{-5}
Train batch size	4
Eval batch size	8
Warmup steps	200
Weight decay	0.01
Max input length	512 tokens
Max output length	128 tokens
Best checkpoint metric	ROUGE-L
Mixed precision	FP16 (GPU)
Estimated training steps	11,046

Table 2: Fine-tuning hyperparameters for BART on SAMSum.

2.4 Evaluation Protocol

All models are evaluated using ROUGE [15] (Recall-Oriented Understudy for Gisting Evaluation), the standard metric for summarization benchmarks. I report F1 scores for ROUGE-1 (unigram overlap), ROUGE-2 (bigram overlap), and ROUGE-L (longest common subsequence), all computed with stemming enabled via the rouge-score library. Evaluation is performed on the first 200 samples of the test set for reproducibility. For inference, BART uses beam search with 4 beams, max_length=128, and min_length=20.

3. Experiments and Results

3.1 Main Quantitative Results

Table 3 presents the ROUGE scores for both models on the 200-sample test subset. The fine-tuned BART model outperforms TextRank by a large and consistent margin across all three metrics.

Model	ROUGE-1	ROUGE-2	ROUGE-L
TextRank (extractive baseline)	28.72	7.94	22.20

Model	ROUGE-1	ROUGE-2	ROUGE-L
BART fine-tuned (abstractive)	51.03	25.80	41.04
Improvement (Δ)	+22.31	+17.86	+18.84
Relative improvement (%)	+77.7%	+224.9%	+84.9%

Table 3: ROUGE-F1 scores (%) on 200 test samples. Higher is better.

Figure 2 provides a visual comparison of these scores as grouped bar charts.

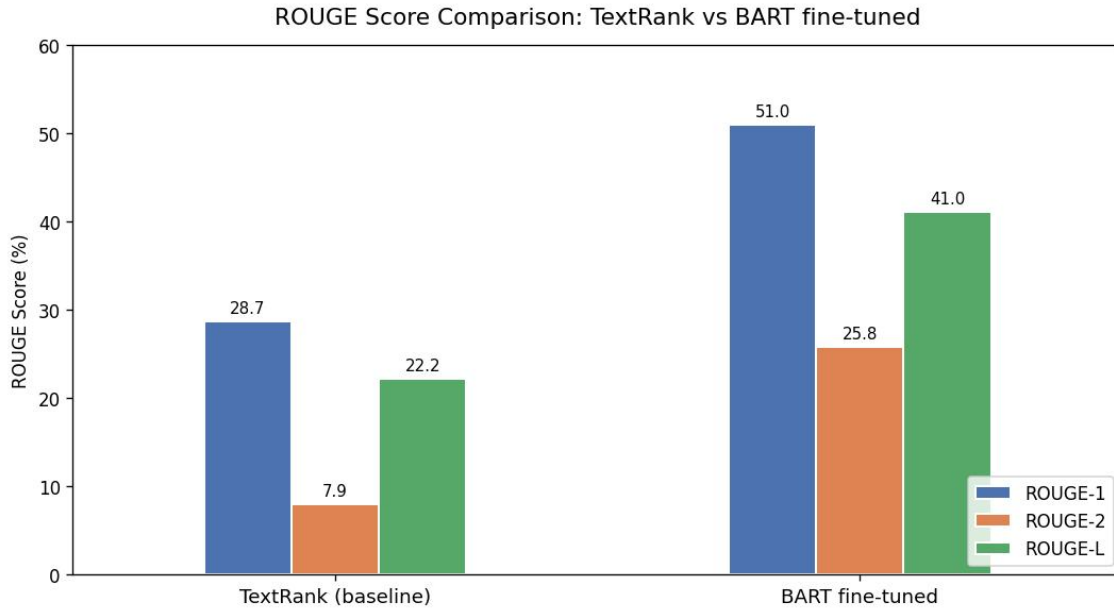


Figure 2: ROUGE-1, ROUGE-2, and ROUGE-L comparison between TextRank and BART fine-tuned.

3.2 Analysis of Results

Several observations follow from Table 3:

- ROUGE-2 shows the largest relative gain (+224.9%, a 3.25× improvement). This metric captures bigram sequence overlap and is particularly sensitive to fluency and coherent phrasing. TextRank's low ROUGE-2 (7.94) reflects that copied dialogue sentences rarely share bigrams with the human-written reference summaries, which use paraphrased, third-person prose. BART's 25.80 indicates substantially closer alignment with reference phrasing.
- ROUGE-L improvement (+18.84) reflects that BART generates longer correct subsequences matching the reference. The gap confirms that BART synthesizes information across multiple dialogue turns rather than selecting individual sentences.
- TextRank ROUGE-1 (28.72) is non-trivial because individual words from dialogue sentences do appear in summaries. The large gap to BART (51.03) demonstrates that word-level recall alone is insufficient — coherent ordering and paraphrasing matter significantly.

The performance of BART (51.03 / 25.80 / 41.04) is consistent with published results for BART-based models on SAMSum, which typically report ROUGE-1 in the range of 50–53, validating

the correctness of our fine-tuning pipeline.

3.3 Per-sample Score Distribution

Figure 3 shows the per-sample ROUGE-L distribution for both models across all 200 test samples. The BART distribution is shifted markedly to the right relative to TextRank and has lower variance, indicating more consistent performance. TextRank has a heavy left tail of near-zero scores on short dialogues where sentence selection produces fragments that share almost no text with the reference summary.

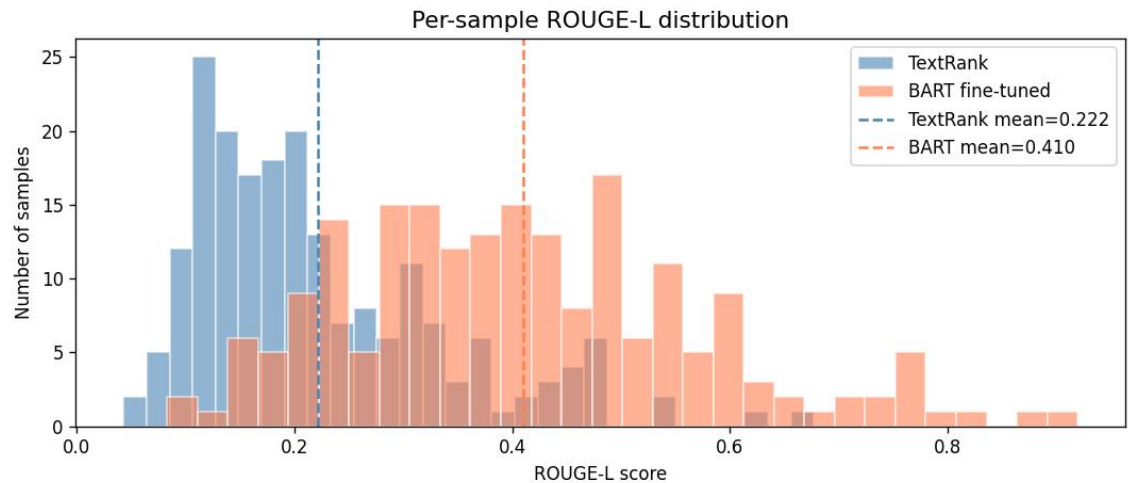


Figure 3: Per-sample ROUGE-L score distributions for TextRank (blue) and BART fine-tuned (coral). Dashed lines indicate means.

4. Case Study and Qualitative Analysis

4.1 Typical Case: Multi-participant Coordination Dialogue (Case #5)

The following example from the test set illustrates a typical multi-participant coordination dialogue, one of the most common dialogue types in SAMSum. The dialogue involves four speakers coordinating key handover and lunch plans.

Field	Content
Dialogue	Benjamin: Hey guys, what are I doing with the keys today? Hilary: I've got them. Whoever wants them can meet me at lunchtime or after Elliot: I'm ok. We're meeting for the drinks in the evening anyway... Benjamin: Hmm I might actually pass by at lunchtime, take the keys and go take a nap. Hilary: Sounds good. We'll be having lunch with some French people at La Cantina at 2pm.
Reference	Hilary has the keys to the apartment. Benjamin wants to get them and go take a nap. Hilary is having lunch with some French people at La Cantina. They're meeting at the entrance to the conference hall at 2 pm.
TextRank	Hilary: Yeah, I guess so Daniel: I'm with Hilary atm and won't let go of her for the

Field	Content
	rest of the day, so any option you guys choose is good for me Benjamin: Hmm I might actually pass by at lunchtime, take the keys and go take a nap.
BART fine-tuned	Hilary has got the keys. She will meet with some French people at the entrance to the conference hall at 2 pm and then they will go to La Cantina for Italian cuisine. Benjamin is going to take a nap.

Table 4: Case #5 — typical multi-participant coordination dialogue.

TextRank selects three verbatim sentences including an uninformative one ("Yeah, I guess so"), producing an incoherent extract. BART correctly identifies the three key facts — who has the keys, the meeting time and place, and Benjamin's plan — and presents them in fluent third-person prose that closely matches the reference. A minor hallucination occurs ("Italian cuisine") which does not appear in the reference, illustrating the tradeoff between fluency and strict faithfulness in abstractive models.

4.2 Best Case: BART Most Outperforms TextRank (Case #70)

Error analysis identified Case #70 as the sample where BART most outperforms TextRank (largest positive ROUGE-L gap). This is a short, simple dialogue about a lost wallet.

Field	Content
Dialogue	Ali: I think I left my wallet at your place yesterday. Could you check? Mohammad: Give me a sec, I'll have a look around my room. Ali: OK. Mohammad: Found it! Ali: Phew, I don't know what I'd do if it wasn't there. Can you bring it to uni tomorrow? Mohammad: Sure thing.
Reference	Ali left his wallet at Mohammad's place. Mohammad'll bring it to uni tomorrow.
TextRank	Mohammad: Give me a sec, I'll have a look around my room. Ali: Phew, I don't know what I'd do if it wasn't there.
BART fine-tuned	Ali left his wallet at Mohammad's place yesterday. Mohammad found it. Mohammad will bring it to the university tomorrow.

Table 5: Case #70 — best case where BART most outperforms TextRank.

TextRank selects two sentences that describe the process of searching rather than the outcome, completely missing the resolution. BART correctly captures the key facts: the lost wallet, its recovery, and the plan to return it. BART's output is nearly identical in content to the reference, demonstrating ideal abstractive summarization behavior on simple factual dialogues.

4.3 Failure Case: BART Underperforms TextRank (Case #140)

Case #140 is the sample where BART most underperforms TextRank — a three-line dialogue about oversleeping.

Field	Content
Dialogue	Sean: I overslept :/ Sam: Again?? Sean: I know.
Reference	Sean overslept again.
TextRank	Sean: I overslept :/ Sam: Again?? Sean: I know.
BART fine-tuned	Sean overslept again and Sam is angry with him for it. He's not happy.

Table 6: Case #140 — failure case where BART underperforms TextRank.

The reference is a single three-word sentence. TextRank scores reasonably because it copies all three utterances verbatim, retaining the key word "overslept." BART, however, adds invented emotional context ("Sam is angry with him," "He's not happy") that does not appear in the dialogue. This is a clear hallucination: the model over-generates by inferring plausible but unsupported emotional states. This failure mode is characteristic of abstractive models on very short inputs where there is insufficient context to constrain generation. In such cases, the extractive baseline is paradoxically more faithful.

4.4 Summary of Qualitative Findings

My case study analysis yields the following insights:

- BART excels at multi-turn coordination dialogues where information is distributed across speakers and turns, producing fluent paraphrased summaries that extractive methods cannot match.
- BART performs best on medium-length dialogues (50–200 words) where context is sufficient to constrain generation but truncation is not an issue.
- BART is prone to hallucination on very short dialogues (under 20 words), where sparse context leads the model to fill gaps with plausible but unsupported inferences.
- TextRank fails systematically on dialogues where no single sentence is self-contained, which is true of most conversational exchanges. Its selections are often fragments that lack coherence when removed from context.

5. Author Contributions

This project was completed independently by a single author. All components of the work were carried out by the same individual, including the literature review and background survey, dataset loading and exploratory data analysis, implementation of the TextRank extractive baseline, BART fine-tuning pipeline setup and training, ROUGE evaluation framework and quantitative analysis, case study and error analysis, and manuscript writing and editing.

6. References

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